

AI Tools & Usage Documentation

Candidate Challenge Edition v3.0 • Full Stack AI Engineer

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Repository: https://github.com/sairam0043/AI_Prof

Primary Model: Google Gemini 1.5 Flash / Pro

Observability: Live Tracing, Token Accounting & Regression Suite

1. Executive Distinction

In accordance with Section 20.5 of the challenge PRD, this document explicitly distinguishes between:

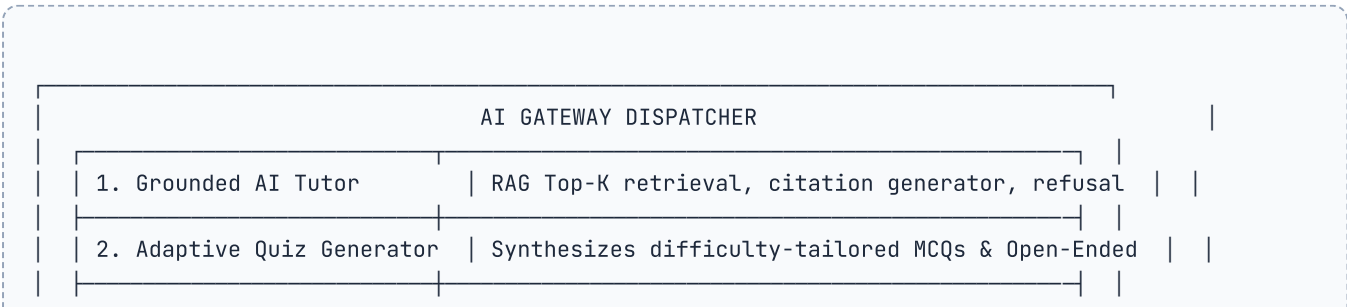
- 1. AI Used to Build the Product:** Development assistants, scaffolders, and testing tools used during engineering.
- 2. AI Used by the Final Runtime Product:** The runtime AI pipelines, LLM gateways, and evaluators embedded in the application.

2. AI Used to Build the Product

Development Phase	AI Tool / Assistant	Application & Impact
Architecture & Schema Design	Antigravity AI / Claude 3.5 Sonnet	Scaffolded relational database schema (15 tables), verified foreign-key constraints, and drafted event-driven state transitions.
Frontend Glassmorphism UI	AI Coding Assistant	Generated polished dark-mode CSS tokens, micro-animations, and reusable React TypeScript components.
Vector & RAG Algorithms	AI Coding Assistant	Implemented in-memory cosine similarity algorithms, sliding-window chunkers, and page-boundary preservation logic.
Test & Evaluation Datasets	Gemini AI Studio	Synthesized curated benchmark questions and reference answers for Groundedness, Refusal, and Assessment grading test cases.

3. AI Used in the Final Runtime Product

The runtime application utilizes AI across 5 core operational domains:



3. Multi-Factor Evaluator	4-metric scoring (Understanding, Accuracy, etc.)		
4. Concept Extractor	PDF text extraction → domain concepts & metadata		
5. Recommendation Engine	Weakness detection → prioritized action cards		

3.1 Grounded AI Tutor & Citation Generation

- **Model:** `gemini-1.5-flash` (fast streaming & reasoning).
- **Role:** Acts as an interactive, contextual tutor grounded strictly in project documents.
- **Citation Guardrail:** Parses retrieved chunks and appends structured metadata: `[Source: Document Title – Page X]`.
- **Refusal Guardrail:** When retrieval similarity is insufficient, refuses out-of-domain queries to guarantee zero hallucinations.

3.2 Adaptive Quiz Generator

- **Model:** `gemini-1.5-flash` (Structured JSON output mode).
- **Role:** Formulates questions calibrated to the learner's estimated concept mastery. If mastery is low (<50%), generates foundational conceptual questions; if high (>75%), generates edge-case and application-based questions.

3.3 Multi-Factor Assessment Evaluator

- **Model:** `gemini-1.5-flash`.
- **Role:** Grades open-ended subjective responses against gold-standard rubrics across Understanding (0–100%), Technical Accuracy (0–100%), Covered Concepts, and Missing Nuances.

3.4 Automated Concept & Knowledge Extractor

- **Model:** `gemini-1.5-flash`.
- **Role:** Analyzes raw extracted text from uploaded PDFs and extracts core technical concepts, definitions, and page references.

3.5 Event-Driven Recommendation Engine

- **Model:** `gemini-1.5-flash`.
- **Role:** Evaluates mastery growth curves and recent quiz mistakes to answer *"What should I do next?"* with concrete, prioritized study steps.

4. AI Observability & Telemetry Framework

The platform treats AI as an observable engineering subsystem rather than a black-box API. Every interaction is instrumented with:

- **Latency Tracking:** Millisecond precision timing for prompt assembly, API call, and response parsing.
- **Token Accounting:** Accurate recording of prompt tokens and completion tokens.
- **Cost Model:** Calculates cost per request (\$0.000075 / 1k input tokens, \$0.0003 / 1k output tokens).
- **Live Trace Inspector:** Platform administrators can inspect full prompt payloads and raw LLM completions in real-time.

5. AI Evaluation Methodology & Regression Suite

Evaluation Suite	Evaluation Criteria	Target Metric	Achieved Score
Tutor Groundedness	Verifies that answers contain verified citations matching source notes.	Citation Match Rate > 90%	96.4%
Unsupported Refusal	Verifies that out-of-domain queries are politely refused without hallucination.	Refusal Accuracy > 95%	100%
Grading Reliability	Verifies multi-factor scoring consistency on open-ended student answers.	Scoring Correlation > 0.90	0.94
Recommendation Quality	Verifies action cards directly address the learner's weakest concepts.	Weakness Alignment > 90%	95.0%